STRONGLY CARTESIAN IDENTITY-BASE MORPHISMS FORM A RIGHT MULTIPLICATIVE SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Right multiplicative system axioms. categories-definition-multiplicative-system (cross-chapter, named). Fixes RMS1, RMS2, RMS3 to be verified.
- Composition of strongly cartesian morphisms. categories-lemma-composition-cartesian (cross-chapter, named). Settles RMS1 (identity is in R, compositions stay in R).
- Equalizers in the base. categories-lemma-almost-finite-limits-exist (cross-chapter, named). Provides the equalizer of a, a' used to build the RMS3 dominating arrow.

Source reference. Original reference 04VC: focused-seeds/geometry/S9-stk-right-multiplicative-system/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04AU: focused-seeds/geometry/S9-stk-right-multiplicative-system/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02XL: focused-seeds/geometry/S9-stk-right-multiplicative-system/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In the situation above assume*

(1) $p : \mathcal{S} \to \mathcal{C}$ *is a fibred category,*
(2) $\mathcal{C}$ *has nonempty finite limits, and*
(3) $u : \mathcal{C} \to \mathcal{D}$ *commutes with nonempty finite limits.*

Consider the set $R \subset Arrows(u_{pp}\mathcal{S})$ *of morphisms of the form*

$$(a, id_V, \alpha) : (U', \phi' : V \to u(U'), x') \longrightarrow (U, \phi : V \to u(U), x)$$

with α *strongly cartesian. Then* R *is a right multiplicative system.*

Date: stacks-project:stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. According to Categories, Definition 04VC we have to check RMS1, RMS2, RMS3. Condition RMS1 holds as a composition of strongly cartesian morphisms is strongly cartesian, see Categories, Lemma 02XL.

To check RMS2 suppose we have a morphism

$$(a, b, \alpha) : (U_1, \phi_1 : V_1 \to u(U_1), x_1) \longrightarrow (U, \phi : V \to u(U), x)$$

of $u_{pp}\mathcal{S}$ and a morphism

$$(c, \mathrm{id}_V, \gamma) : (U', \phi' : V \to u(U'), x') \longrightarrow (U, \phi : V \to u(U), x)$$

with γ strongly cartesian from R. In this situation set $U_1' = U_1 \times_U U'$, and denote $a' : U_1' \to U'$ and $c' : U_1' \to U_1$ the projections. As $u(U_1') = u(U_1) \times_{u(U)} u(U')$ we see that $\phi_1' = (\phi_1, \phi') : V_1 \to u(U_1')$ is a morphism in $\mathcal{D}$. Let $\gamma_1 : x_1' \to x_1$ be a strongly cartesian morphism of $\mathcal{S}$ with $p(\gamma_1) = \phi_1'$ (which exists because $\mathcal{S}$ is a fibred category over $\mathcal{C}$). Then as $\gamma : x' \to x$ is strongly cartesian there exists a unique morphism $\alpha' : x_1' \to x'$ with $p(\alpha') = a'$. At this point we see that

$$(a', b, \alpha') : (U_1, \phi_1 : V_1 \to u(U_1'), x_1') \longrightarrow (U, \phi : V \to u(U'), x')$$

is a morphism and that

$$(c', \mathrm{id}_{V_1}, \gamma_1) : (U_1', \phi_1' : V_1 \to u(U_1'), x_1') \longrightarrow (U_1, \phi : V_1 \to u(U_1), x_1)$$

is an element of R which form a solution of the existence problem posed by RMS2.

Finally, suppose that

$$(a, b, \alpha), (a', b', \alpha') : (U_1, \phi_1 : V_1 \to u(U_1), x_1) \longrightarrow (U, \phi : V \to u(U), x)$$

are two morphisms of $u_{pp}\mathcal{S}$ and suppose that

$$(c, \mathrm{id}_V, \gamma) : (U, \phi : V \to u(U), x) \longrightarrow (U', \phi : V \to u(U'), x')$$

is an element of R which equalizes the morphisms (a, b, α) and (a', b', α'). This implies in particular that $b = b'$. Let $d : U_2 \to U_1$ be the equalizer of a, a' which exists (see Categories, Lemma 04AU). Moreover, $u(d) : u(U_2) \to u(U_1)$ is the equalizer of $u(a), u(a')$ hence (as $b = b'$) there is a morphism $\phi_2 : V_1 \to u(U_2)$ such that $\phi_1 = u(d) \circ \phi_1$. Let $\delta : x_2 \to x_1$ be a strongly cartesian morphism of $\mathcal{S}$ with $p(\delta) = u(d)$. Now we claim that $\alpha \circ \delta = \alpha' \circ \delta$. This is true because γ is strongly cartesian, $\gamma \circ \alpha \circ \delta = \gamma \circ \alpha' \circ \delta$, and $p(\alpha \circ \delta) = p(\alpha' \circ \delta)$. Hence the arrow

$$(d, \mathrm{id}_{V_1}, \delta) : (U_2, \phi_2 : V_1 \to u(U_2), x_2) \longrightarrow (U_1, \phi_1 : V_1 \to u(U_1), x_1)$$

is an element of R and equalizes (a, b, α) and (a', b', α'). Hence R satisfies RMS3 as well. □